19th May, 2025 (Tue)

M2T1 scheme of evaluation

① a) Write the postulates of QFET. [3]

b) What is Fermi-Dirac distribution function and explain its variation at $T=0K$ & $T>0K$ [3]

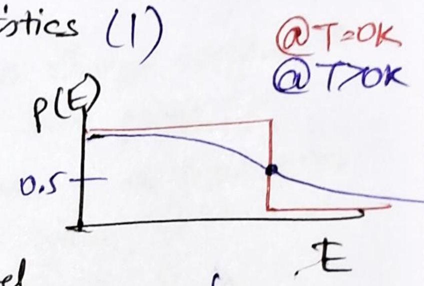
c) Calculate temperature at which there is 3% probability of finding electron with energy >1eV above Fermi level? [4]

a) Postulates

- Wave nature of electron (1)
- Pauli's exclusion principle (1)
- Fermi Dirac dist statistics (1)

b) $$P_{FD}(E) = \frac{1}{1+\exp\left(\frac{E-E_F}{k_BT}\right)}$$

(1)



(2)

E = energy of level

E_F = Fermi-energy level

k_B = Boltzmann's constant

T = temperature

$P_{FD}(E)$ = probability of occupying level E

c) $E = E_F + 1eV$ (1)

$P_{occupy}(E) = P_{FD}(E) = 0.03$ (1)

$$0.03 = \frac{1}{1+\exp\left(\frac{E-E_F}{k_BT}\right)} = \frac{1}{1+\exp\left(\frac{1eV}{k_BT}\right)} \simeq 100K$$

$$\exp\left(\frac{1eV}{k_BT}\right) = \frac{0.97}{0.03}$$

$$T = \frac{8.63}{26} \times 300$$

(2)

$$\frac{1eV}{k_BT} = 115.86 \Rightarrow k_BT = \frac{1000meV}{115.86} = 8.63meV$$

② a) What is the principle of solar cell? How light energy is converted into electrical energy using solar cell? [3]

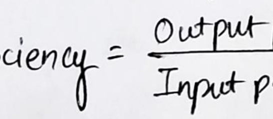
b) Draw I-V- characteristics and obtain an expression of efficiency and fill factor. [3]

c) A solar cell is having an area of 1cm x 1cm and input light power density is 1000W/m². The efficiency is 5%. Calculate load resistance value to draw maximum output power when current of 5mA is flowing in the circuit. [4]

a) - Photo-voltaic effect.
- Working principle
  - Generation of excess charge carriers (1)
  - Separation of electron hole pairs (1)
  - ~~Generation~~ Accumulation of electrons at n region & holes at p-region. (1)

b)

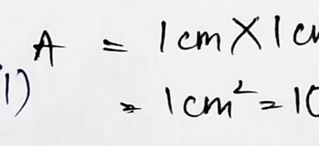
I



V

(1)

c)



$I_{ir} = 1000 W/m^2$

$A = 1cm \times 1cm$
$= 1cm^2 = 10^{-4} m^2$

$\eta = 5\% = 0.05$

Input power $= I_{ir} \times A$
$= 10^3 \cdot 10^{-4} W$
$= 10^{-1} W$ (1)

$\Rightarrow$ Output power $= \eta \cdot P_{in}$

$P_{out} = 5 \cdot 10^{-3} W$

Output current $= 5mA$ (1)

$\Rightarrow P_{\cancel{out}} = I^2 R \Rightarrow R = \frac{5mW}{25 m \cdot m} = 200\Omega$ (2)

(η) Efficiency $= \frac{\text{Output power}}{\text{Input power}} \times 100$ (1)

(FF) Fill factor $= \frac{\text{Output power}}{\text{Ideal power}}$ (1)

$\eta = \frac{V_{mp} I_{mp}}{I_{ir} \times A} \times 100$, I_{ir} = Input irradiance.

$FF = \frac{V_{mp} \times I_{mp}}{V_{oc} \cdot I_{sc}}$

(3) (a) How the numerical aperture and acceptance angle are related with each other and draw the mathematical relation between them. [2]

* Numerical aperture (N.A.) is defined as the sine of acceptance angle (θ_0)

$$N.A. = \sin\theta_0 \quad (1)$$

but $\theta_0 = \sin^{-1}\left(\sqrt{n_1^2 - n_2^2}\right)$

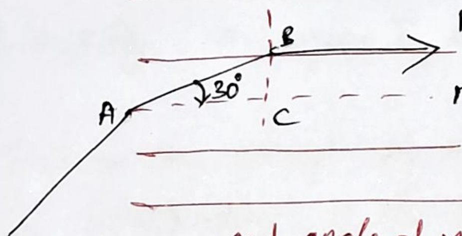
where n_1 = refractive index of core

n_2 = refractive index of cladding

$$\therefore N.A. = \sin\left(\sin^{-1}\sqrt{n_1^2 - n_2^2}\right) \quad (1)$$

$$N.A. = \sqrt{n_1^2 - n_2^2}$$

(b) Light ray is launching from air to core with index 1.5



$n_2 = ?$

$n_1 = 1.5$

$\sin\theta_0 = 0.75$

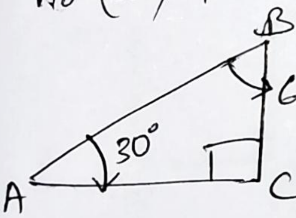
$\theta = 48.60^\circ$

and angle of refraction is 30°. Calculate

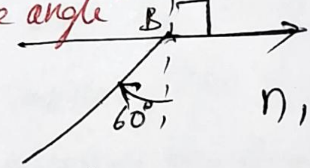
(i) Refractive index of cladding n_2 [4]

(ii) Acceptance angle

n_0 (air) = 1



$\Rightarrow$



$\angle B = \frac{\pi}{2} - \angle A$

$= 60^\circ$

(i) From Snell's law

$$\frac{\sin 60^\circ}{\sin 90^\circ} = \frac{n_2}{n_1}$$

$$\frac{\sqrt{3}}{2} = \frac{n_2}{n_1}$$

$$\Rightarrow n_2 = \frac{\sqrt{3}}{2} n_1 = 1.3 \quad (2)$$

(ii) Acceptance angle (θ_0)

from Snell's law

$$\frac{\sin\theta_0}{\sin 30^\circ} = \frac{n_1}{n_0} \quad (2)$$

$\Rightarrow \sin\theta_0 = \sin 30 \cdot n_1$

(c) Also calculate the numerical aperture (N.A.) and fractional refractive index (Δ) of optical fiber [4]

N.A. = ? and Δ = ?

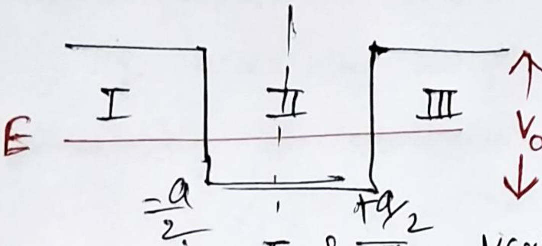
$N.A. = \sin\theta_0$

$= 0.75 \quad (2)$

$N.A. = n_1\sqrt{2\Delta}$

$\Rightarrow \Delta = \frac{1}{2}\left(\frac{N.A.}{n_1}\right)^2 = 0.125 \quad (2)$

4 (a) An electron is confined between the two boundaries of height V_0 as shown in below figure. The energy of the particle is less than the height of the barrier ($E < V_0$). Apply the Schrodinger equation in regions I & III and find the probability. [4 M]



Schrodinger equation

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$$

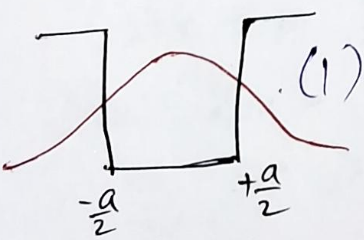
In regions I & III, $V(x) = V_0$, so that

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\psi = 0 \quad (2)$$

Probability in region III is $\int_{+\frac{a}{2}}^{\infty} |\psi|^2(x)\, dx$ (1)

Probability in region I is $\int_{-\infty}^{-\frac{a}{2}} |\psi|^2(x)\, dx.$ (1)

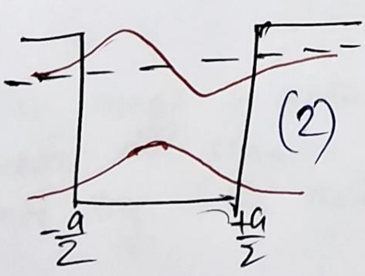
(b) Explain the concept of tunneling by plotting the wavefunctions of the finite potential well. [2]



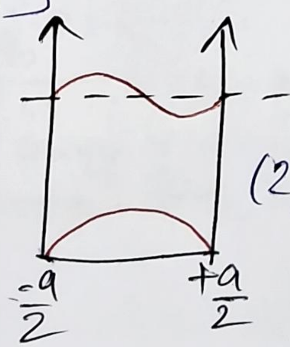
(1)

The wavefunctions "penetrates"/ "tunnels" into the finite potential barrier. This is called quantum tunneling (1)

(c) Compare the wavefunctions of the finite and infinite potential wells. [4]



(2)



(2)

5 a) Write a note on Heisenberg uncertainty principle and its significance [2M]

The Heisenberg uncertainty principle states that the position and momentum of a particle cannot measured simultaneously. (1)

If uncertainty in position is Δx and uncertainty in momentum is Δp, then

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2} \quad (1)$$

b) Prove that a free electron cannot exist inside a nucleus using the Heisenberg uncertainty principle. [4M]

Let us prove this by method of contradiction.

Assume the electron can exist inside nucleus.

The radius of nucleus is of ~~the~~ order 10^{-15} m

Then by Heisenberg uncertainty principle

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Here $\Delta x = 10^{-15}$ m

$$\Rightarrow \Delta p \geq \frac{\hbar}{2\Delta x} = \frac{6.625 \cdot 10^{-34}}{2 \cdot 2 \cdot 3.14 \cdot 10^{-15}} \quad (2)$$

$$= 5.27 \cdot 10^{-20}\ \text{kg m s}^{-1}.$$

So energy of electron =

$$\frac{(\Delta p)^2}{2m} = \frac{(5.27 \cdot 10^{-20})^2}{2 \cdot 9.1 \cdot 10^{-31}}\ \text{J} \quad (2)$$

$$\text{In eV} = \frac{(5.27)^2 \cdot 10^{-40}}{2.9.1 \cdot 10^{-31}\ 1.6 \cdot 10^{-19}}\ \text{eV} = 9.53 \cdot 10^{9}\ \text{eV}$$

This energy is greater than energy of the nucleus. Therefore ~~this~~ situation is impossible. Hence electron cannot stay inside nucleus

(5)(C) Electrons moving with a velocity of $3.32 \cdot 10^{5} ms^{-1}$ if this velocity is measured with an inaccuracy of 1%, then estimate the uncertainty in the position of an electron. [4 M]

$$\frac{\Delta v}{v} = 1\% = 0.01 \Rightarrow \Delta v = 0.01 \cdot v$$

$$\Rightarrow \frac{\Delta p}{p} = \frac{\Delta (mv)}{p} = \frac{m(\Delta v)}{mv} = \frac{\Delta v}{v} = 0.01$$

$$\therefore \Delta p = m \Delta v \quad (2)$$

Now Δx is given by

$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

$$\Delta x \geq \frac{\hbar}{2 \Delta p}$$

$$= \frac{\hbar}{2 m \Delta v}$$

$$= \frac{6.625 \cdot 10^{-34}}{2 \cdot 2 \cdot 3.14 \cdot 9.1 \cdot 10^{-31} \cdot 0.01 \cdot 3.32 \cdot 10^{5}}$$

$$= \frac{6.625 \cdot 10^{-1-5}}{4 \cdot 3.14 \cdot 9.1 \cdot 3.32}$$

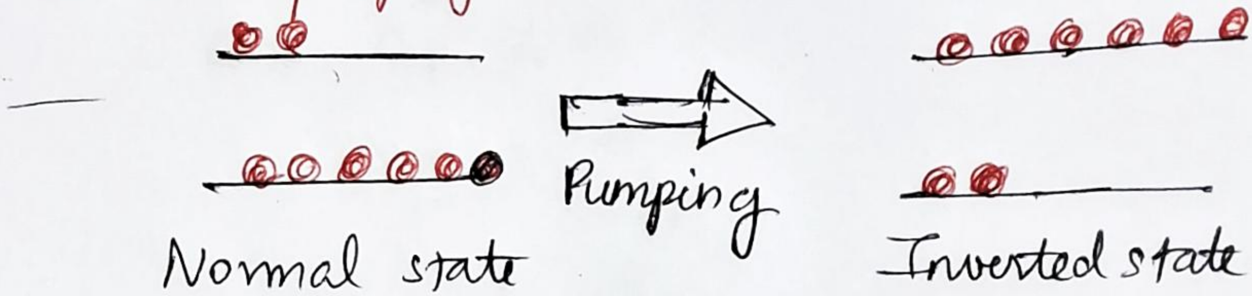
$$\Delta x = 1.74 \cdot 10^{-8} \text{ m} \quad (2)$$

6) a) Outline the characteristic features of lasers. [3]

Laser characteristics

- Unidirectional
- Monochromatic
- Coherent
- High intensity
- Low pulse width

b) Analyse population inversion and various pumping mechanisms in lasers [3]



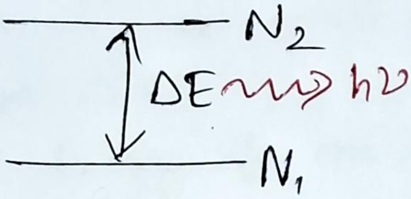
Population inversion is the process of converting normal state into inverted state by means of pumping. (2)

Types of pumping

- Optical pumping
- Electrical pumping

(1)

(6) (c) The wavelength of light emitted from a laser is 610 nm at room temperature. Determine the ratio of population of higher to lower energy state. How the ratio is affected, if temperature changes from room temperature to 400K? [4]



@ thermal equilibrium

$$\frac{N_2}{N_1} = \exp\left(-\frac{\Delta E}{K_B T}\right)$$

At $\lambda = 610\,nm$

energy of photon $\Delta E = \frac{1.24\,eV}{\lambda\,(\mu m)}$

$$= \frac{1.24\,eV}{0.61}$$

$$= 2\,eV$$

So $\frac{N_2}{N_1} = \exp\left(-\frac{2000\,meV}{26\,meV}\right)$

$$= 3.92 \cdot 10^{-34}$$ (2)

@ 400K

$$\frac{N_2}{N_1} = \exp\left(-\frac{2000\,meV}{K_B 400K}\right)$$

$$K_B 400K = K_B 300K \cdot \frac{400}{300} = 26\,meV \cdot \frac{4}{3}$$

$$= 34.67\,meV$$

$\therefore \frac{N_2}{N_1} = \exp\left(-\frac{2000\,meV}{34.67\,meV}\right)$

$$= 8.85 \cdot 10^{-26}$$ (2)

(8) (a) What is meant by molar specific heat? [2]

Molar specific heat is the heat required to raise the temperature of one mole of material by one ~~degree~~ kelvin (1)

$$C = \frac{dE_{1mole}}{dT} \quad (1)$$

(b) Explain how the CFET failed to explain the electronic specific heat. [4]

From CFET.

Energy of one electron at temperature T

$$\Delta E_{1electron} = \frac{3}{2} K_B T \quad (1)$$

⇒ Energy of one mole of electrons at temp T

$$\Delta E_{1mole} = \frac{3}{2} K_B T \times N_A$$

$$= \frac{3}{2} (K_B N_A) T \quad (1)$$

But $K_B N_A = R$ // Ideal gas constant

$$\therefore \Delta E_{1mole} = \frac{3}{2} RT$$

$$\therefore C = \frac{dE_{1mole}}{dT} = \frac{\Delta E_{1mole}}{\Delta T} = \frac{3}{2} R \quad (1)$$

Since $R = 8.314 \text{ J mol}^{-1} K^{-1}$

$$C = \frac{3}{2} \times 8.314 \approx 12 \text{ J mol}^{-1} K^{-1} \quad (1)$$

However, experimentally observed electronic contribution molar specific heat is two orders of magnitude lower.

(7)(C) Elucidate how the QFET is successful in explaining electronic specific heat by ~~over~~ overcoming the failures of CFET. [4]

In QFET, the fraction of electrons participating in thermalization process is

$$\frac{K_B T}{E_F} \quad (1)$$

So change in energy per 1 electron at temperature T is

$$\Delta E_{1\,electron} = (K_B T)\left(\frac{K_B T}{E_F}\right) \quad (1)$$

∴ The change in energy per 1mole of electrons at temperature T is

$$\Delta E_{1\,mole} = \frac{(K_B T^2)}{E_F} K_B N_A$$

$$= \left(\frac{K_B R}{E_F}\right) T^2 \quad (1)$$

∴ electronic part of molar specific heat

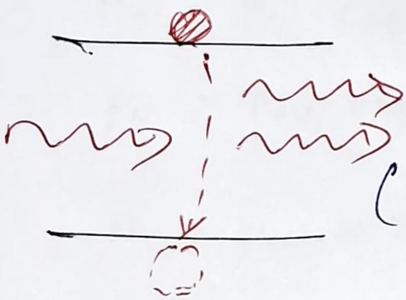
$$C = \frac{\Delta E_{1mole}}{\Delta T} = \frac{K_B R}{E_F} \cdot 2T$$

$$= 2\left(\frac{K_B T}{E_F}\right) \cdot R \quad (1)$$

@ Room temperature $\frac{K_B T}{E_F} \simeq 0.01$

∴ $C \simeq 0.01 R$ in agreement with experiment

(8) (a) What is meant by stimulated emission. Explain the necessity of stimulated emission in obtaining the properties of LASER. [3M]



(1) In the stimulated emission, an incoming photon stimulates / triggers electron in higher energy level to transition to lower energy level and result in emission of another photon with same energy (1)

$$h\nu + \text{electron}^* \longrightarrow \text{electron} + 2h\nu$$

Necessity of stimulated emission :)

(b) Explicate the essential conditions of LASER? [3M]

There are three conditions of LASER –

High photon density

Large lifetime of excited level (1)

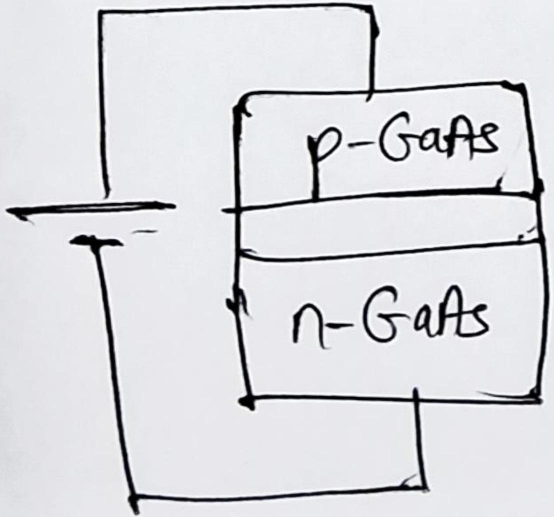
High population of excited level

Population inversion (1)

Metastable ~~state~~ level (1)

Radiation confinement (1)

(8) c) Elucidate the working of semiconductor LASER with the help of band diagram [4]



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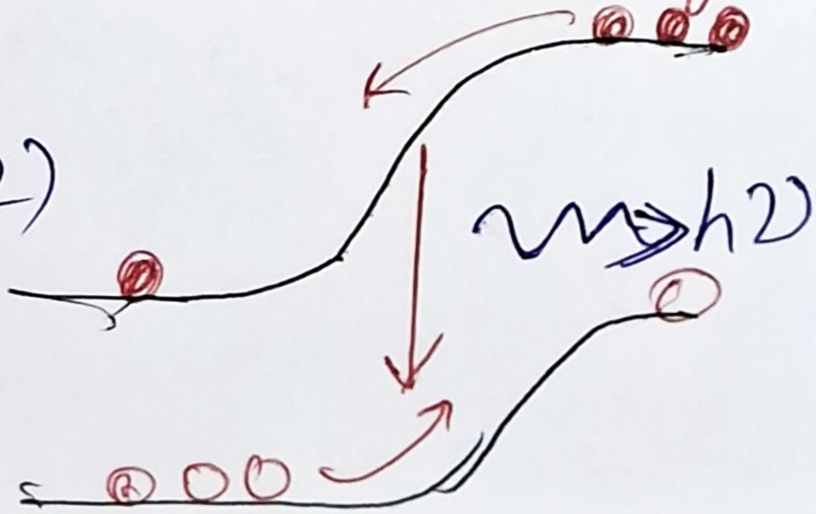


(2)

(9) a) Starting from the expressions of electron and hole concentrations, obtain an expression for the electrical conductivity of an intrinsic semiconductor. [4M]

$$n_o = N_c \exp\left(-\frac{E_c - E_F}{K_B T}\right)$$

$$p_o = N_v \exp\left(-\frac{E_F - E_v}{K_B T}\right)$$

$$n_o = p_o = n_i$$

and $n_o p_o = n_i^2 = N_c N_v \exp\left(-\frac{E_c - E_v}{K_B T}\right)$

$$= N_c N_v \exp\left(-\frac{E_g}{K_B T}\right)$$

(2) $\Rightarrow n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2K_B T}\right)$

Now conductivity in intrinsic S.C.

$$\sigma_i = \sigma_e + \sigma_h$$

$$= n_o e \mu_e + p_o e \mu_h \quad (1)$$

But $n_o = p_o = n_i$

$$\Rightarrow \sigma_i = n_i e (\mu_e + \mu_h)$$

$$= \sqrt{N_c N_v}\, e (\mu_e + \mu_h) \exp\left(-\frac{E_g}{2K_B T}\right)$$

$$\sigma_i = \sigma_o \exp\left(-\frac{E_g}{2K_B T}\right) \quad (1)$$

(b) Why Ge has more conductivity than GaAs ~~at~~ all temperatures? [2M]

Band gap of Ge < Band gap of GaAs

$$\Rightarrow \exp\left(-\frac{E_g(Ge)}{2K_BT}\right) > \exp\left(-\frac{E_g(GaAs)}{2K_BT}\right)$$

$$\Rightarrow \quad \sigma(Ge) > \sigma(GaAs)$$

(c) Conductivity of Si is $4.4 \cdot 10^{-4}\,\Omega^{-1}m^{-1}$ at 20°C

Calculate its conductivity at 100°C given the bandgap of Si is 1.1eV. [4M]

$$\frac{\sigma(100°C)}{\sigma(20°C)} = \frac{\sigma(473K)}{\sigma(293K)} = \frac{\exp\left(-\frac{E_g}{K_BT_{473}}\right)}{\exp\left(-\frac{E_g}{K_BT_{293}}\right)}$$

(2)

$$\therefore \sigma(473K) = \exp\left[-\left(\frac{E_g}{K_BT_{473}} - \frac{E_g}{K_BT_{293}}\right)\right] \cdot \sigma(293K)$$

Now $K_BT_{473} = K_BT_{300} \cdot \frac{473}{300} = 41\,meV$

$K_BT_{293} = K_BT_{300} \cdot \frac{293}{300} = 25.4\,meV$

$$\therefore \sigma(473K) = \exp\left[-\left(\frac{1100}{41} - \frac{1100}{25.4}\right)\right] 4.4 \cdot 10^{-4}\,\Omega^{-1}m^{-1}$$

$$= \exp\left(-(26.83 - 43.30)\right) \cdot 4.4 \cdot 10^{-4}\,\Omega^{-1}m^{-1}$$

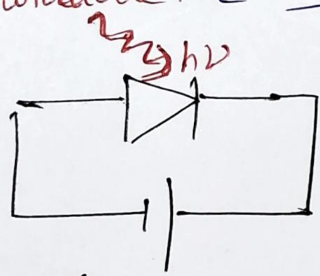
$$= 1.43 \cdot 10^{7} \cdot 4.4 \cdot 10^{-4}\,\Omega^{-1}m^{-1}$$

$$= 6.3 \cdot 10^{+3}\,\Omega^{-1}m^{-1}. \qquad (2)$$

(10) (a) Both solar cell and photodiode convert the light energy into electrical energy. What are the basic differences between them in terms of operation? [3M]

Solar cell	Photo diode
(1) - pn junction @ zero bias	p-n junction @ reverse bias
(1) - works on the basis of photo voltaic effect	works on the basis of photoconductive effect
(1) power is generated	power is consumed.

(b) Describe the working mechanism of photodiode. [3M]

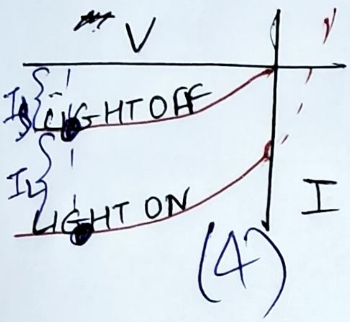


Working principle
- Photo conductive effect (1)
- pn junction is under reverse bias.

Case (i) LIGHT OFF
(1) So a reverse bias current, also called DARK current flows through the photodiode when the LIGHT is not illuminating it

Case (ii) LIGHT ON
(1) Under bright conditions, the current due to excess charge carriers generated causes addition to dark current.

(c) Draw the I-V characteristics of photodiode as a function of incident light intensity. [4M]



@ LIGHT OFF

$I = I_S$ = reverse saturation current/dark current

@ LIGHT ON

$I = I_S + I_L$,

where I_L = photo current.

X

End of PART-A